

AI Zero-Waste Kitchen: Household Food Waste Reduction

A smart cooking assistant for matching leftover ingredients with safe, allergen-free recipes to minimize household food waste.

Primary Target Audience	Students, budget-conscious individuals, and home cooks with food allergies.
Core Scope	Ingredient matching, allergen filtering, recipe generation, and fallback recommendations.
Repository	https://github.com/aviusL/INF1103-G8-Assignment.git

1. Problem Statement and Target Users

Students and home cooks regularly leave ingredients unused because they lack the culinary knowledge to combine their random leftovers into good-tasting and complete meals. Additionally, individuals with food allergies face health risks when improvising dishes without clear dietary boundaries. Existing recipe platforms cater primarily to pre-planned grocery shopping rather than cooking dynamically with on-hand ingredients.

Our main target audience is university students and home cooks that have to manage tight budgets or food allergies. The initial project focuses on dynamic, inventory-constrained meal generation and automated allergen enforcement

Project Objectives

- Reduce avoidable household food waste by maximizing the use of on-hand ingredients.
- Prevent allergen exposure through strict rule-based output verification.
- Provide clear recipe options and minimal shopping recommendations when ingredients fall short.
- Take into account ingredient expiry dates, prioritising the use of ingredients with sooner expiries.

2. User Inputs

Input	Example
Available Ingredients	Eggs, Spinach, Cooked Rice, Soy Sauce
Allergen Exclusions	Peanuts, Dairy, Gluten
Dietary Preferences	Vegetarian
Preparation Time Limit	15 mins
Available Equipment	Microwave, Stovetop

3. Use of AI

Every recipe request passes through the AI engine. The AI then analyses available ingredients, accounting for cooking constraints like allergens or blacklisted recipes, and generates an optimized recipe using available inventory. The AI then returns a structured output. The structured output will include a finalized recipe title, a strictly filtered ingredient list with calculated measurements, step-by-step cooking instructions based on the user's equipment, along with safety and allergen tags.

4. Business Rules

The logic manager combines user inputs and AI-generated outputs. Recipe suggestions are recommendations, not automatic food safety guarantees. Users must confirm ingredient freshness before cooking.

Condition	System Action
AI output contains user-specified allergens	Reject recipe immediately and request regenerated output
All recipe ingredients match user inventory	Present the recipe directly to the user
1-2 minor non-essential ingredients missing	Display recipe alongside a minimal shopping suggestion list
Insufficient ingredients to make a valid dish	Advise user not to cook and recommend purchasing key staples
AI confidence or safety score is low	Place output under fallback review and load a static pre-verified recipe